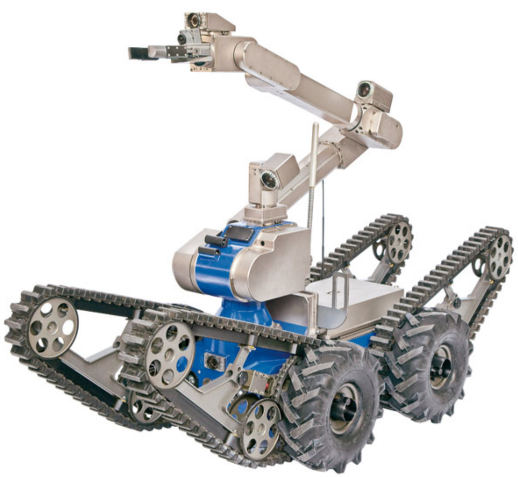
1st WORKSHOP FTEC-2015
SPANISH TRAINEESHIP PROGRAMME

Robotic projects low level control development



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Section: *EN-STI-ECE*
Ref. Number: *EN2734*

CERN | EN ENGINEERING DEPARTMENT

Ciemat
Centro de Investigaciones Energéticas, Medioambientales y Tecnológicas

Train Inspection Monorail (TIM)

Goal

- Development of Unmanned tasks along LHC

Features

- Attached to ceiling
- 5 Wagons
- 3G UMTS / Wi-fi
- LabVIEW GUI

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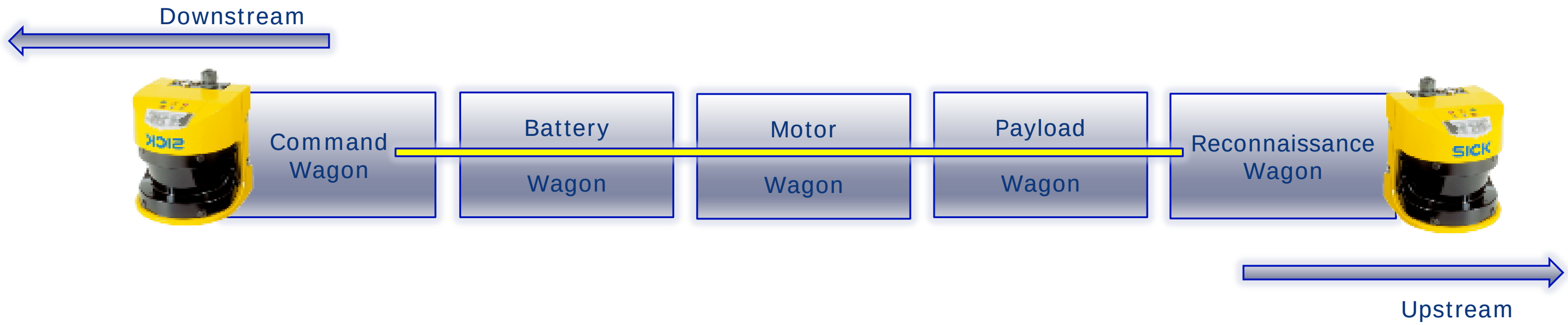

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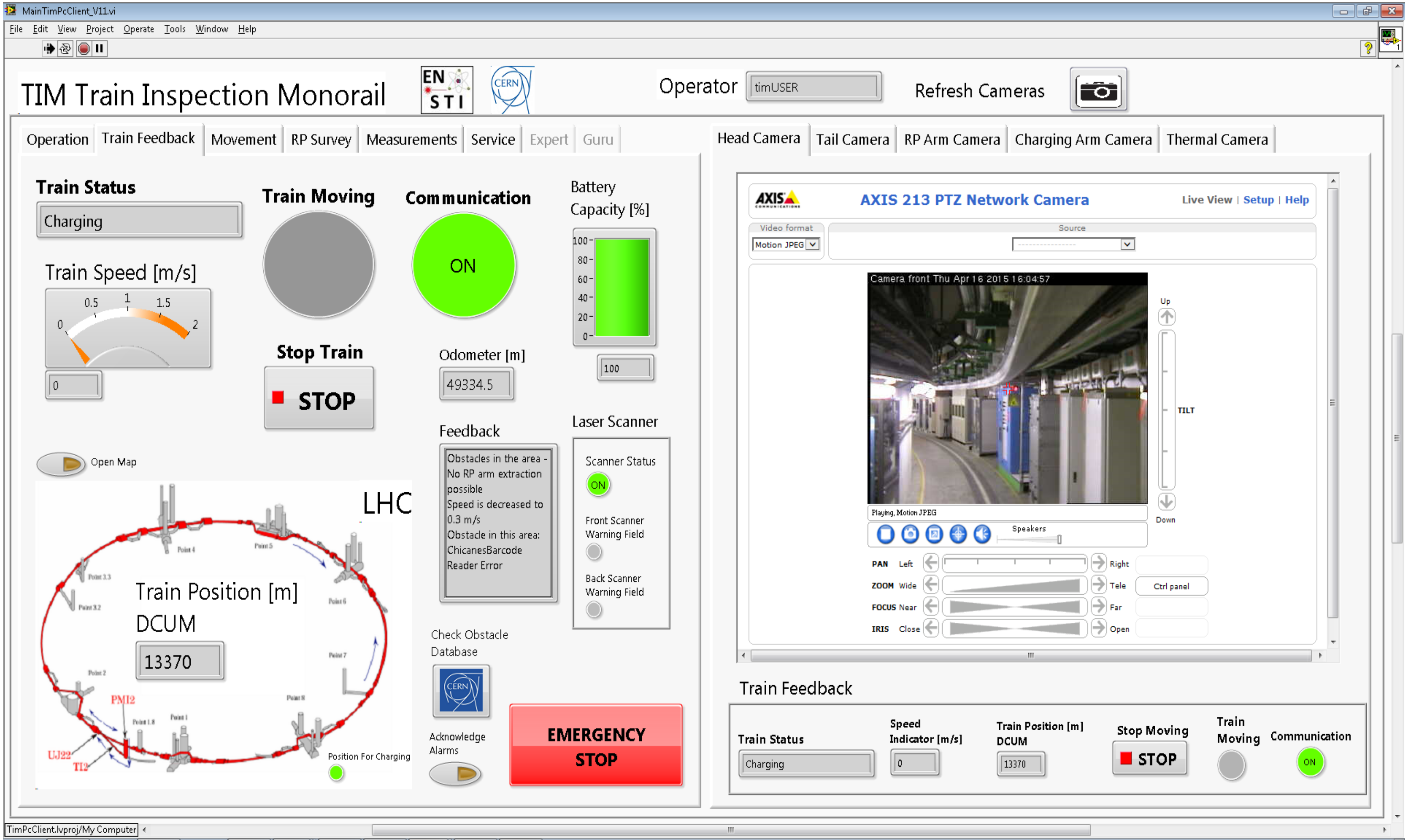

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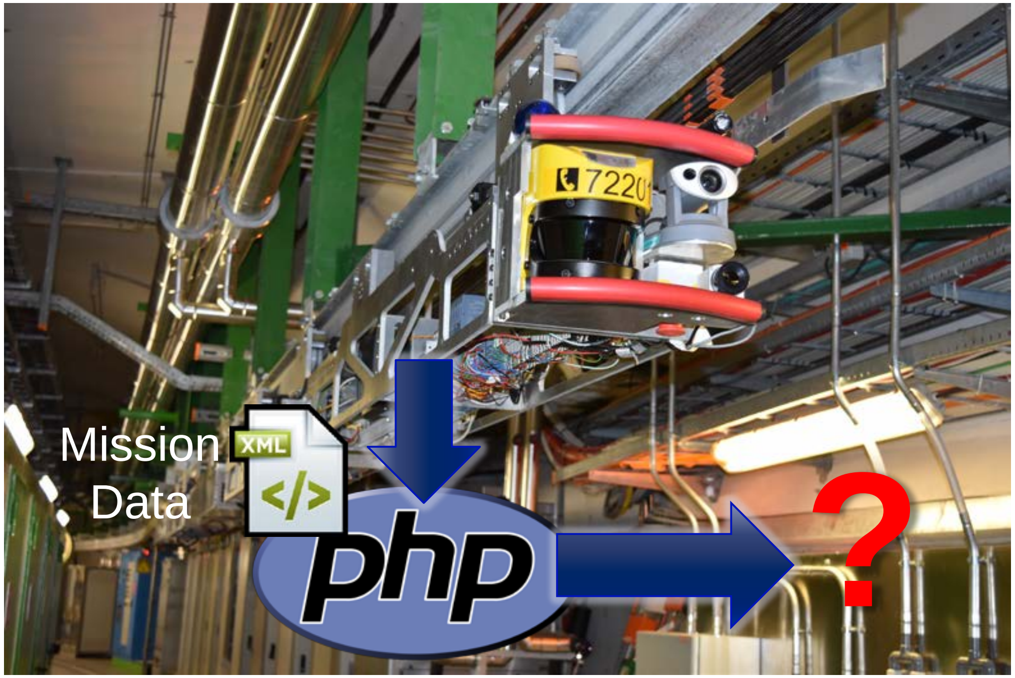

Features

- Attached to ceiling
- 5 Wagons
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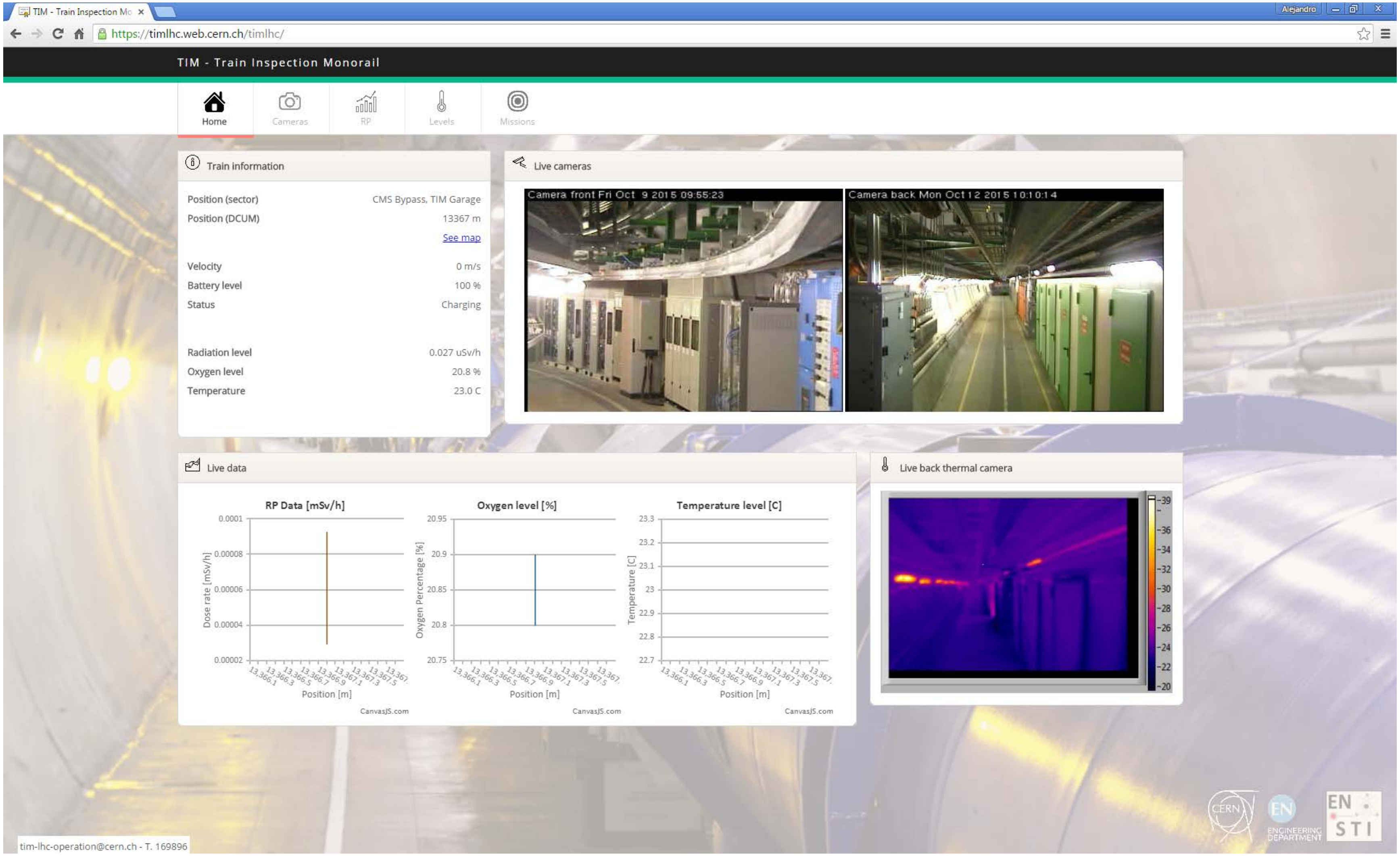
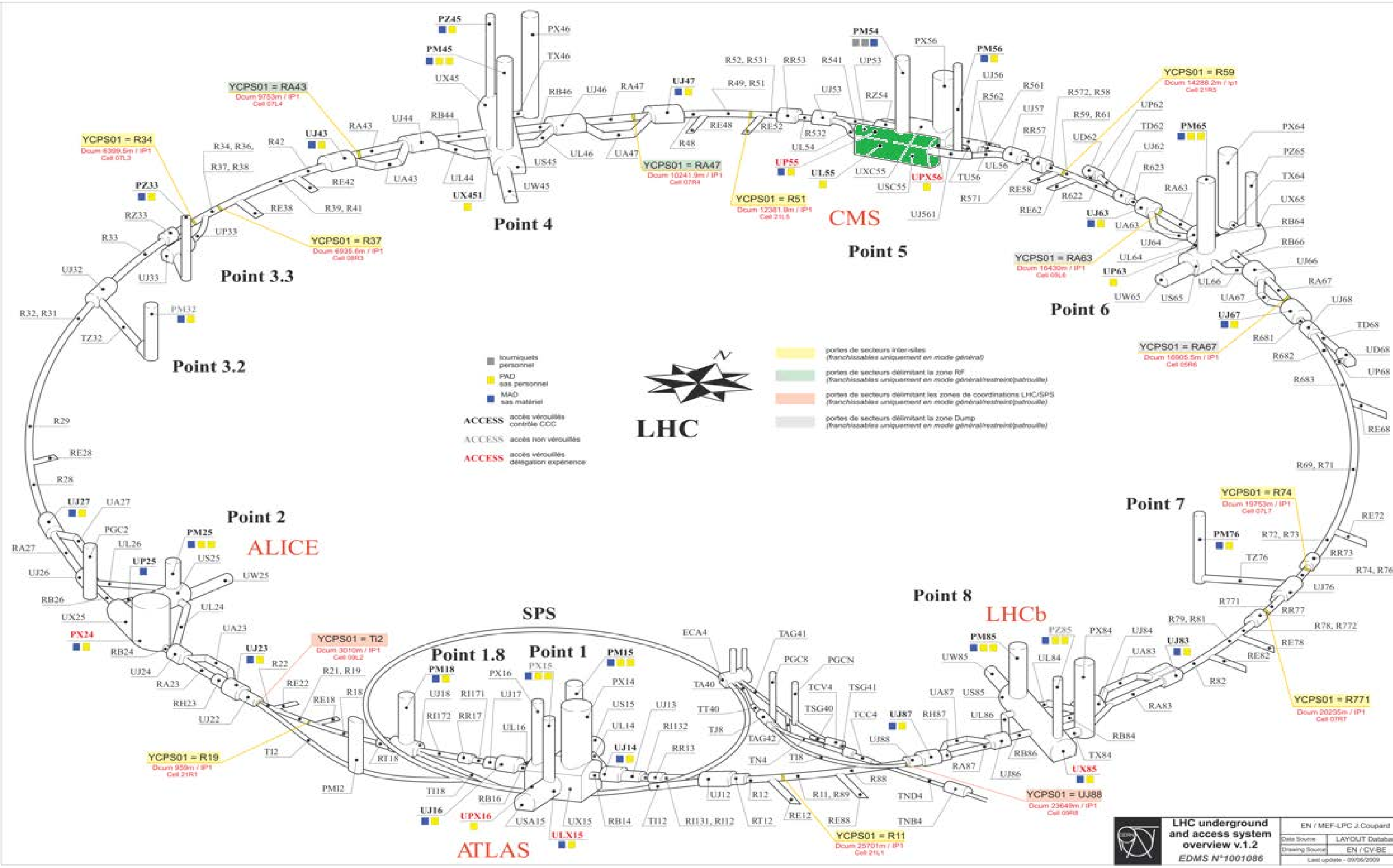
Main Tasks

- Checking of collimators displacements.
- Measuring of RP dose and Temperature, TIM position, images, etc.

Monitoring and Measurements Treatment



Monitoring and Measurements Treatment

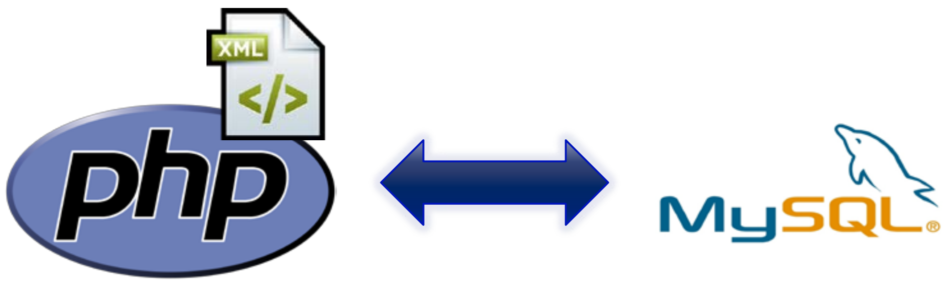


Monitoring and Measurements Treatment

Goals

- Data Management
- Graphical information
- Show images
- Check TIM position

Contribution



Monitoring and Measurements Treatment

Management options

- Insert new Missions in DB
- Queries to the DB
- Show list of the inserted Missions

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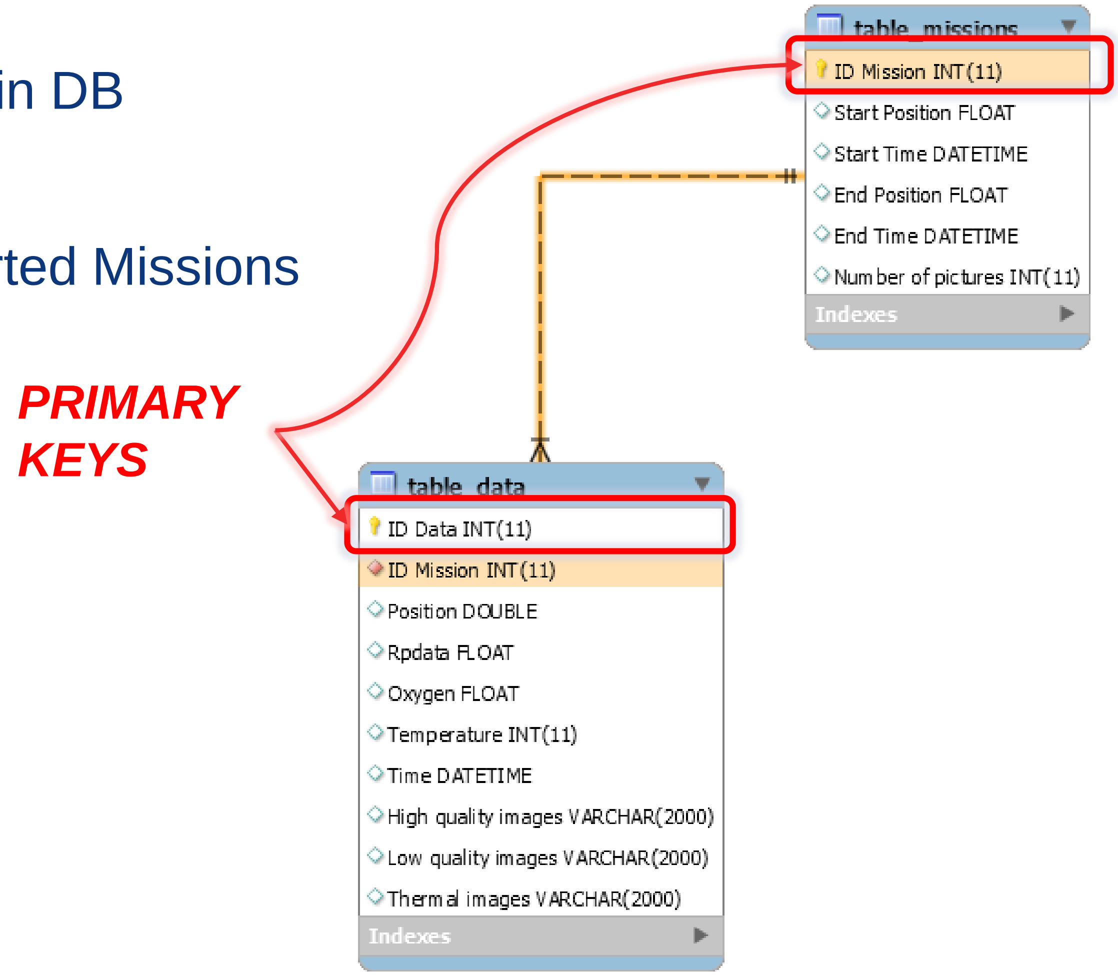
2 Tables

- ***Table Missions***
- ***Table Data***



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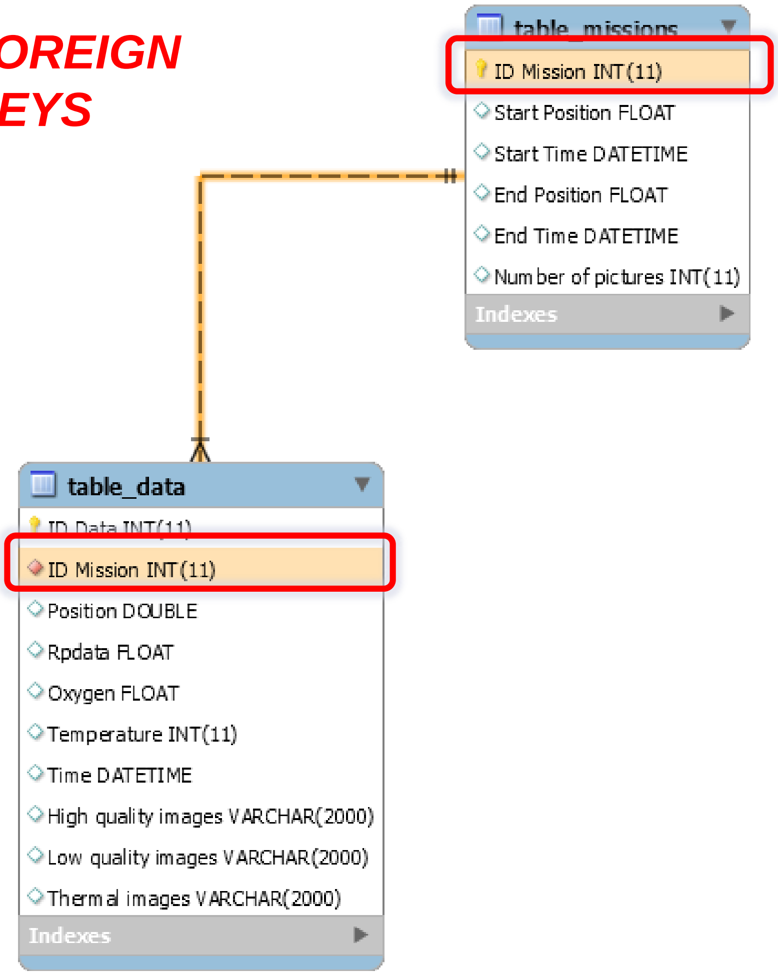
FOREIGN KEYS



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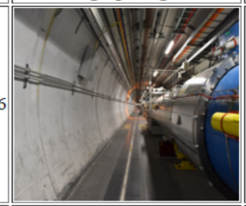
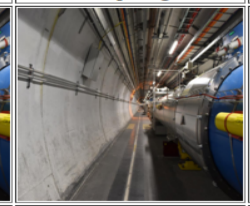
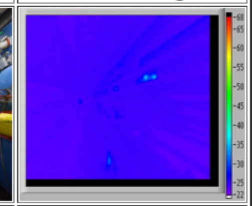
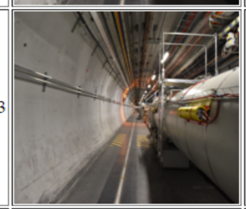
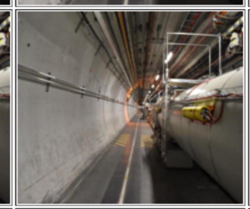
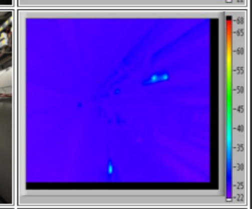
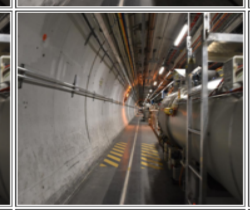
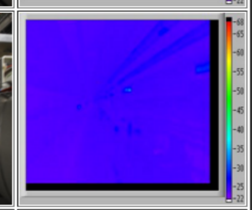
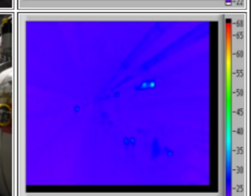
Monitoring and Measurements Treatment

https://timlhc.web.cern.ch/timlhc/model/TIM_DB_rw.php

Connected successfully
Are going to be updated the Missions
The Mission Table and the Data Table were created in the TIM_DB Database
The Mission are going to be updated in the DB

Table 1 Missions	ID Mission	Start Position	Start time	End Position	End time	Number of pictures
	1	13174.96	2015-07-09 12:20:33	12824.92	2015-07-09 12:35:46	157

Table 2 Data	ID Data	ID Mission	Position	Rpdata	Oxygen	Temperature	Time	Highq images	Lowq images	Thermal images
	1	1	13174.96	0.366	21.3	25	2015-07-09-12:20:36			
	2	1	13172.87	0.24	21.3	25	2015-07-09-12:20:53			
	3	1	13170.84	0.208	21.3	25	2015-07-09-12:20:58			
	4	1	13168.74	0.145	21.3	25	2015-07-09-12:21:03			

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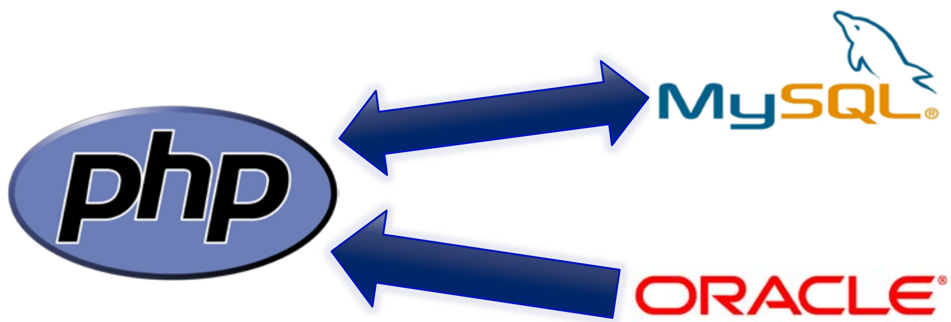

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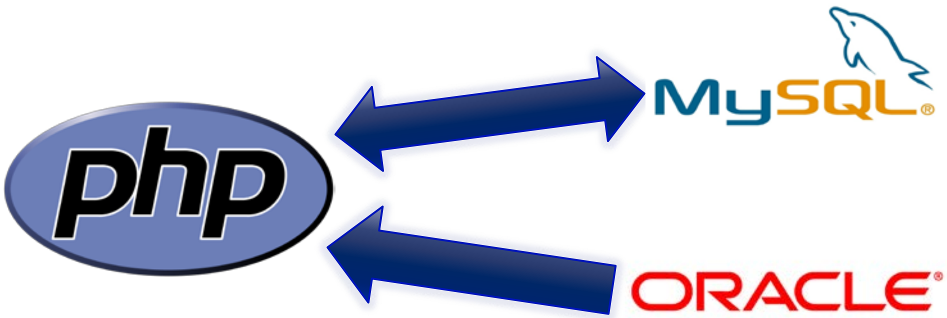

Challenges

- ERROR Uploading Images

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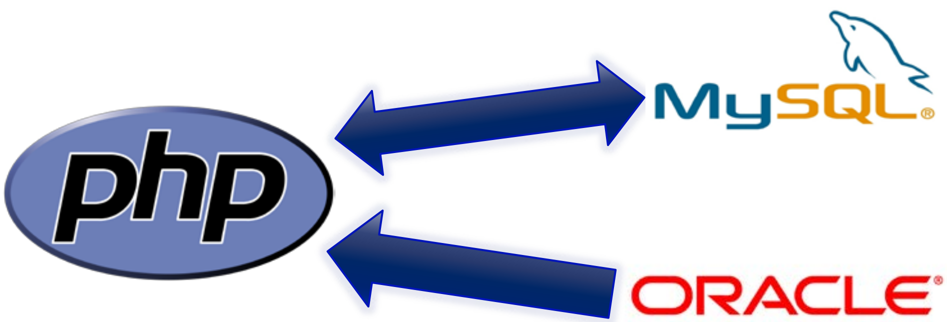

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Status: *Operational*

Telemanipulation force feedback

Goal

- Force feedback in telemanipulated robotic arms

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Starting idea

- Bracelet user + Sensors in gripper + Wireless Comm.

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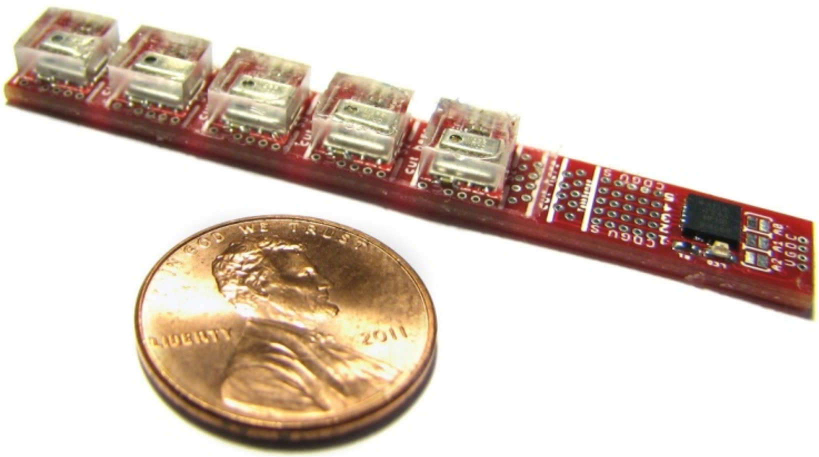

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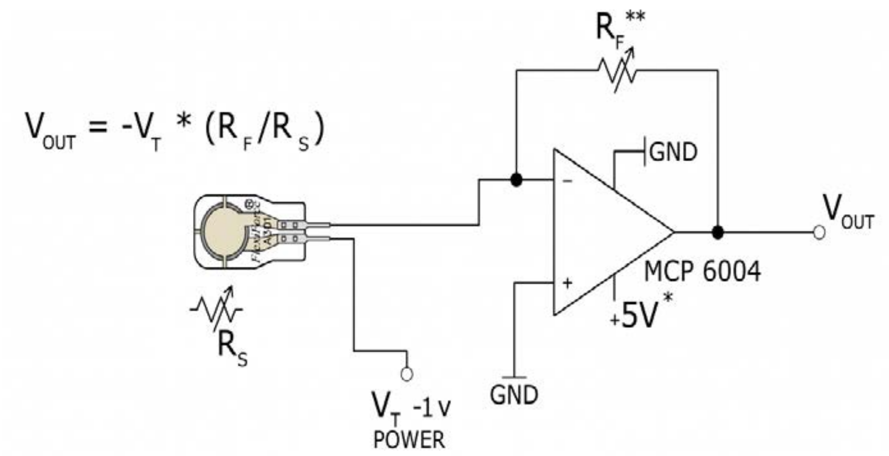

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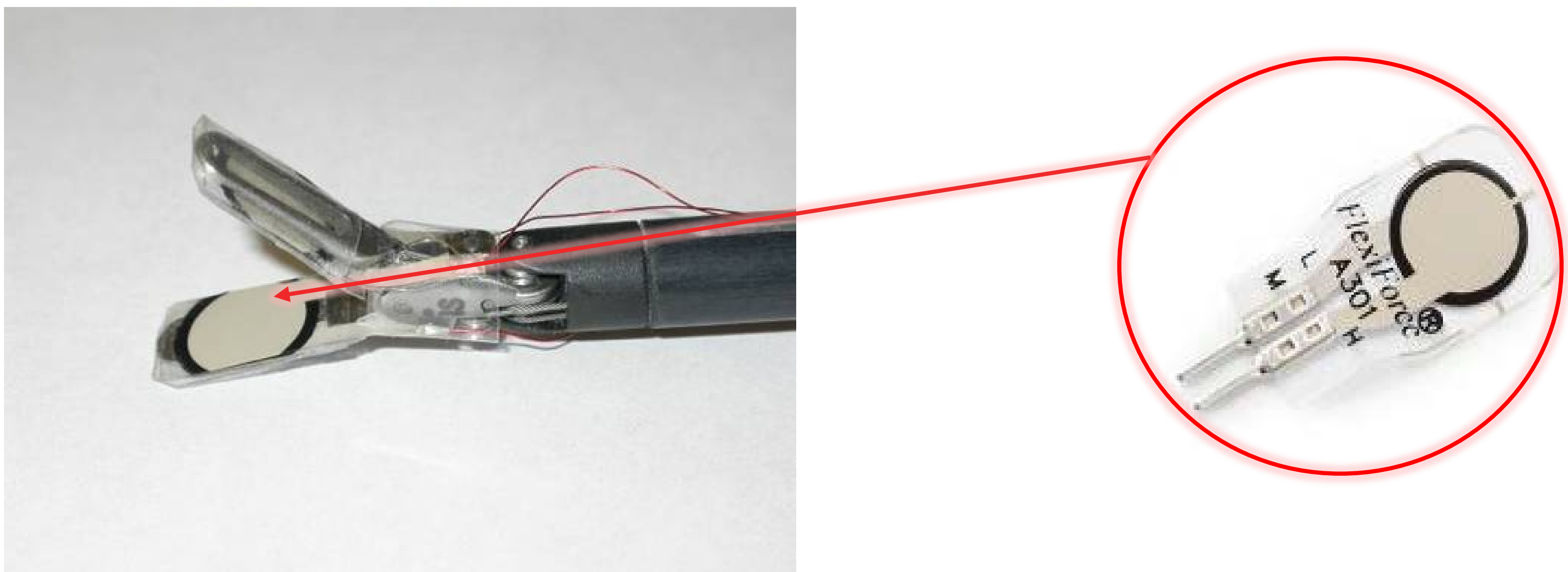

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Status: R&D

Future Guideline

- Continue working on DB and Force Feedback Projects

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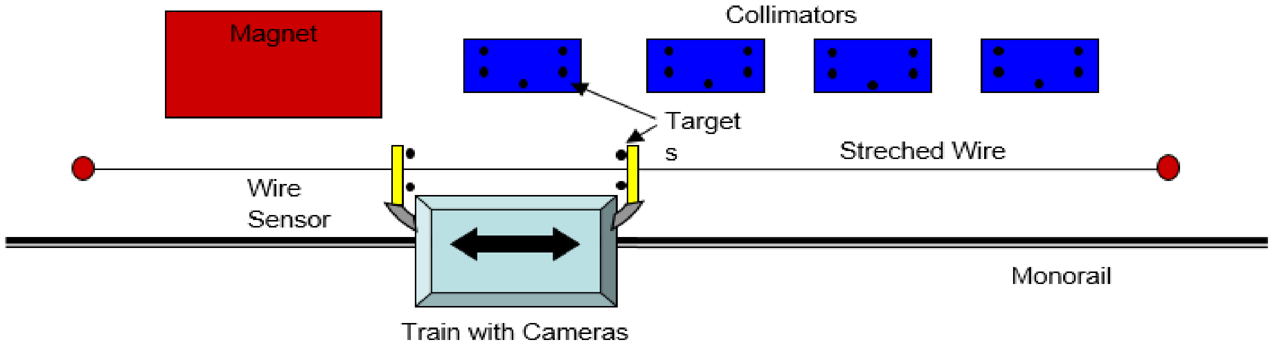
- ➢2 New Projects:
  - ❖Integration of Survey Wagon in TIM.



Future Guideline

- ➢Continue working on DB and Force Feedback Projects

- ➢2 New Projects:
  - ❖Integration of Survey Wagon in TIM.
  - ❖Reverse Engineering of Robotic arm.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